

Loci of complex numbers

Basic skills

Q1. a) $|2+3i| = \sqrt{2^2+3^2} = \sqrt{13}$

$$\text{Arg}(2+3i) = \tan^{-1}\left(\frac{3}{2}\right) = 0.983$$

b) $|3i-2| = \sqrt{3^2+2^2} = \sqrt{13}$

$$\text{Arg}(3i-2) = \tan^{-1}\left(\frac{3}{-2}\right) = -0.983$$



as $x < 0$, $y > 0 \rightarrow +\pi$

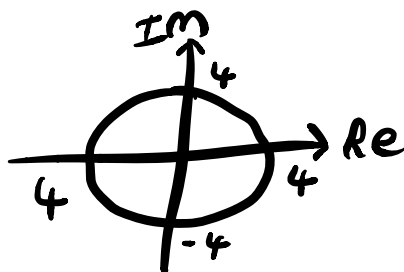
c) $|-(1+i)| = |1+i| = \sqrt{2}$

$$\text{Arg}(-(1+i)) = \tan^{-1}\left(\frac{-1}{-1}\right) = \frac{\pi}{4} - \pi = -\frac{3\pi}{4}$$



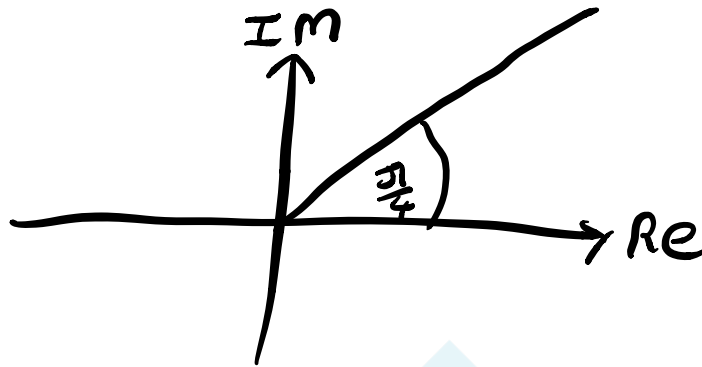
Q2. a) $|z|=4$ circle radius: 4

centre: $0+0i$

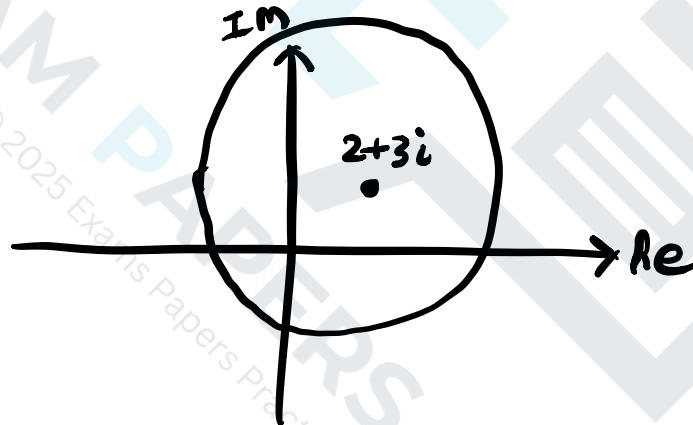


i.e. $\uparrow$
(0,0)

b) $\text{Arg}(z) = \frac{\pi}{4}$ half-line at angle $\frac{\pi}{4}$



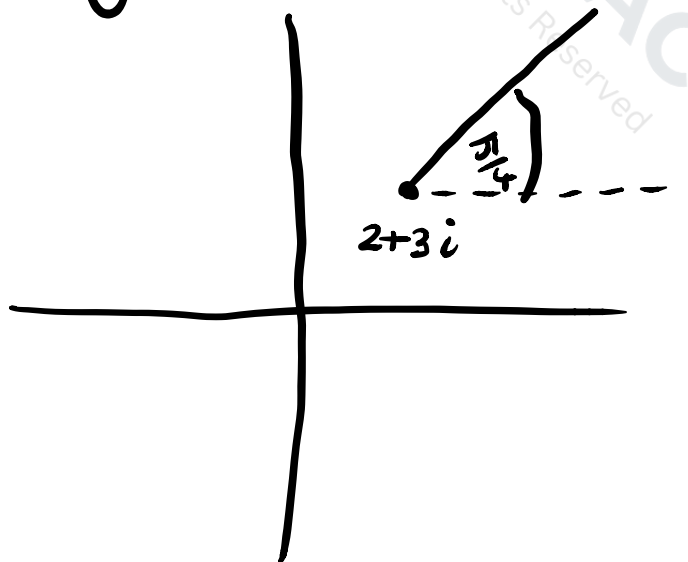
c) $|z - (2+3i)| = 4$ circle radius: 4



centre: $2+3i$

i.e. $\nearrow$
(2, 3)

d) $\text{Arg}(z - (2+3i)) = \frac{\pi}{4}$ halfline at $2+3i$ with angle $\frac{\pi}{4}$ on positive horizon

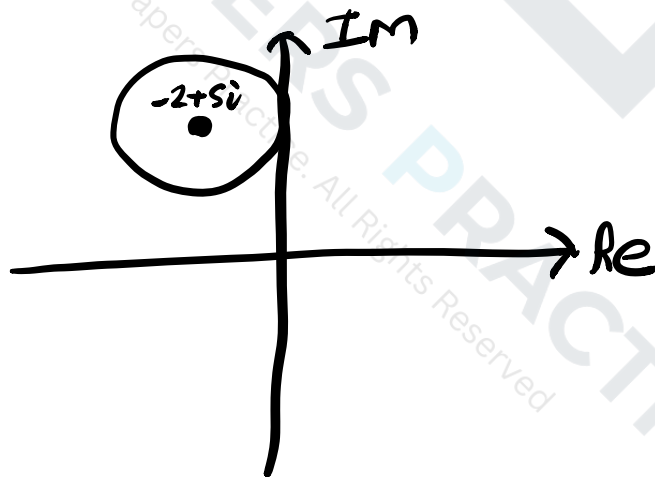


Competency skills

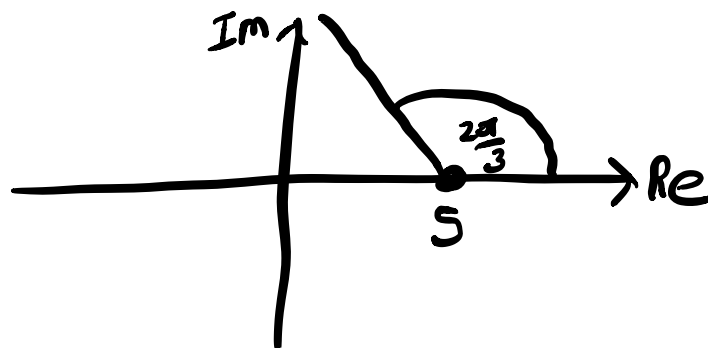
Q3. $|z - 3 + 4i| = |z - (3 - 4i)|$

is the distance from $3 - 4i$ so
centre should be $3 - 4i$ so
wrong as they drew it at $3 + 4i$

Q4. a) $|z - 5i + 2| = 2$
 $|z - (-2 + 5i)| = 2$

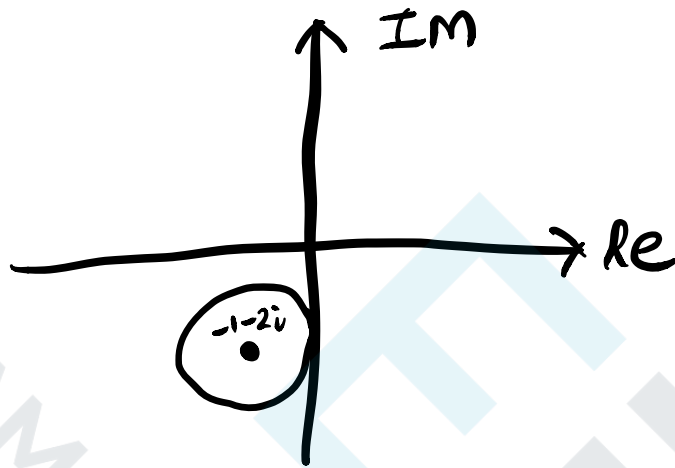


b) $\text{Arg}(z - 5) = \frac{2\pi}{3}$



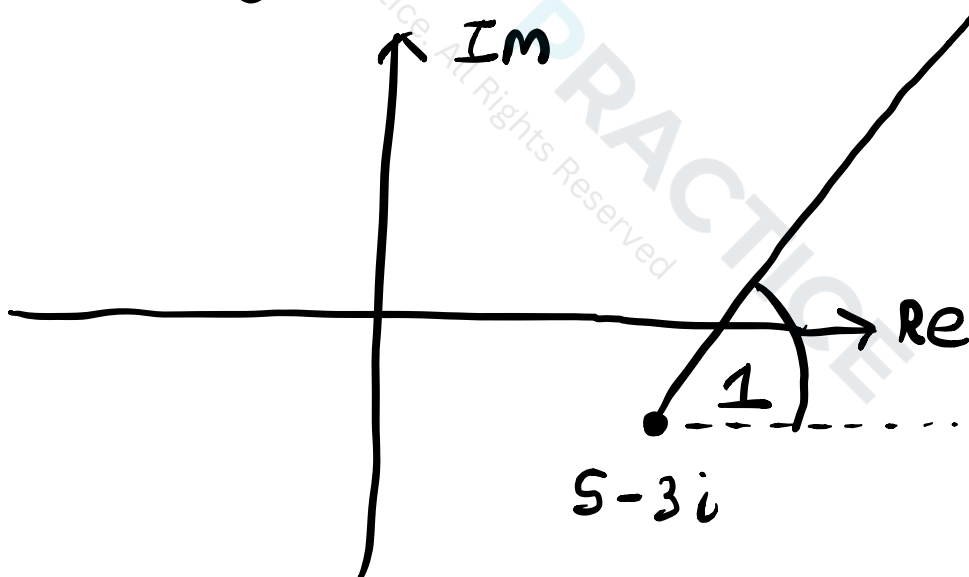
c) $|z + 2i + 1| = 1$

$|z - (-1 - 2i)| = 1$



d) $\text{Arg}(z + 3i - 5) = 1$

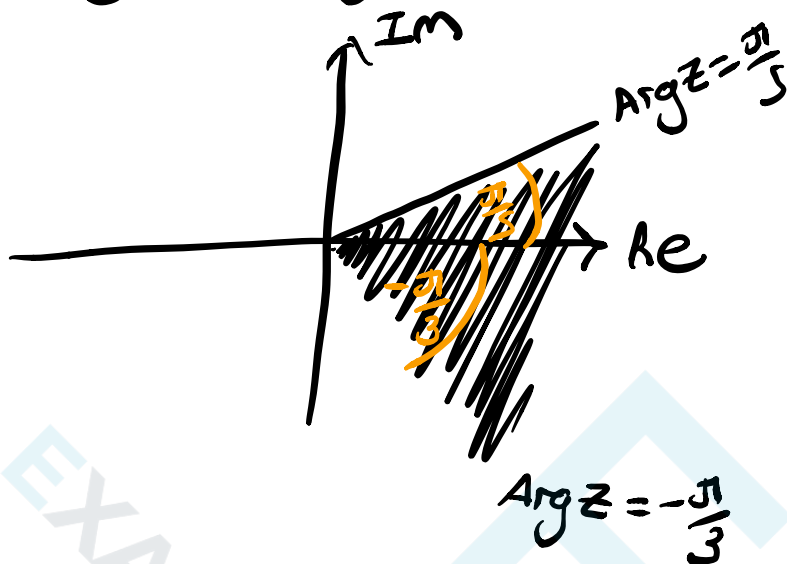
$\text{Arg}(z - (5 - 3i)) = 1$



Q5. $|z| > 1$ draw $|z| = 1$ dotted then outside as distance greater than 1

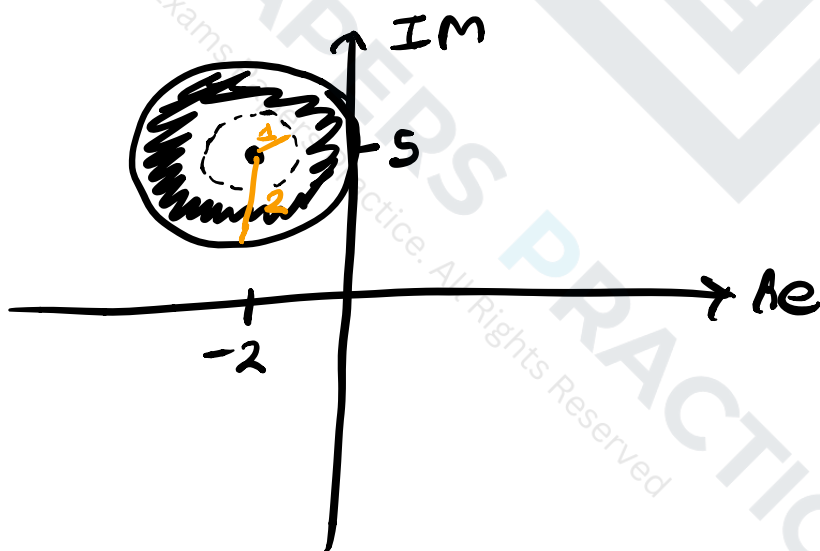


Q6. $-\frac{\pi}{3} < \text{Arg } z \leq \frac{\pi}{5}$ draw $\text{Arg } z = \frac{\pi}{5}$



$\text{Arg } z = -\frac{\pi}{3}$
then any angle
between

Q7. a) $1 < |z - (-2 + 5i)| \leq 2$



b)

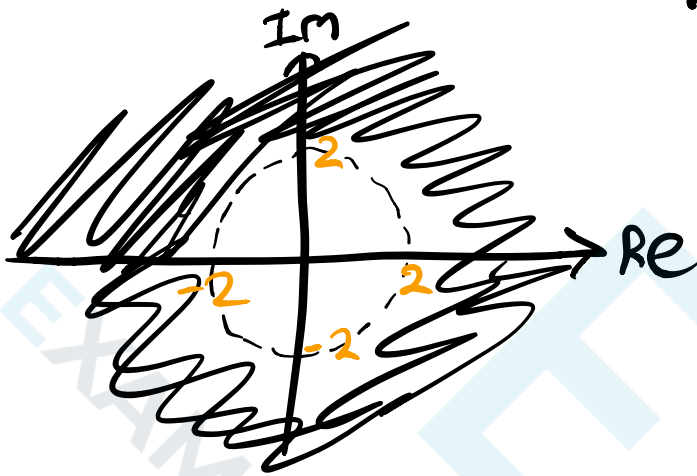
$$\text{Arg } z < \frac{2\pi}{3}$$



c) $|z^2| > 4$ so $|z|^2 > 4$

so

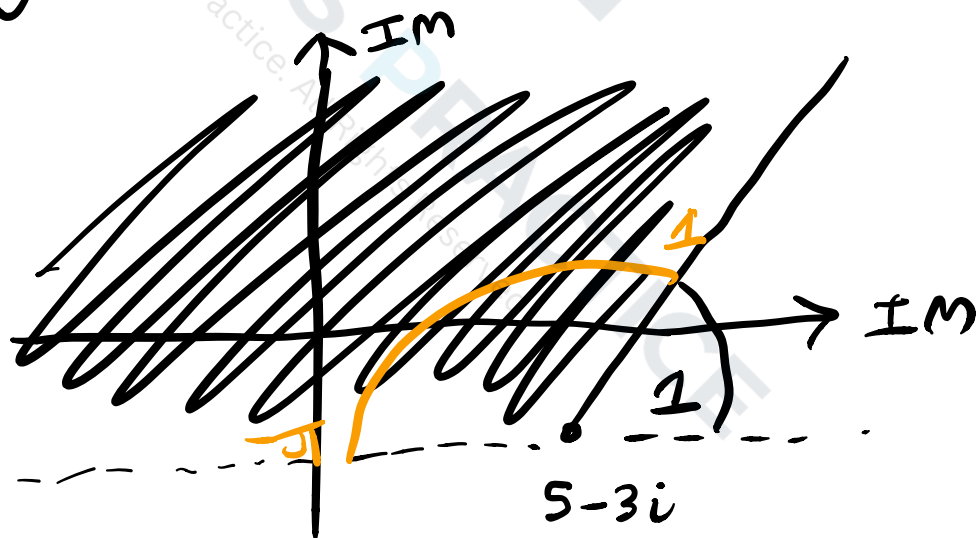
$|z| > 2$ (or $|z| < -2$ impossible though)



d) $\text{Arg}(z + 3i - 5) > 1$

$\pi > \text{Arg}(z - (5 - 3i)) > 1$

↑
upper
limit



Advanced skills

Q 8. a) $z - w = (x - u) + (y - v)i$

$$|z - w| = \sqrt{(x - u)^2 + (y - v)^2}$$

$$w - z = (u - x) + (v - y)i$$

$$|w - z| = \sqrt{(u - x)^2 + (v - y)^2}$$

$$= \sqrt{(x - u)^2 + (y - v)^2}$$

$$= |z - w|$$

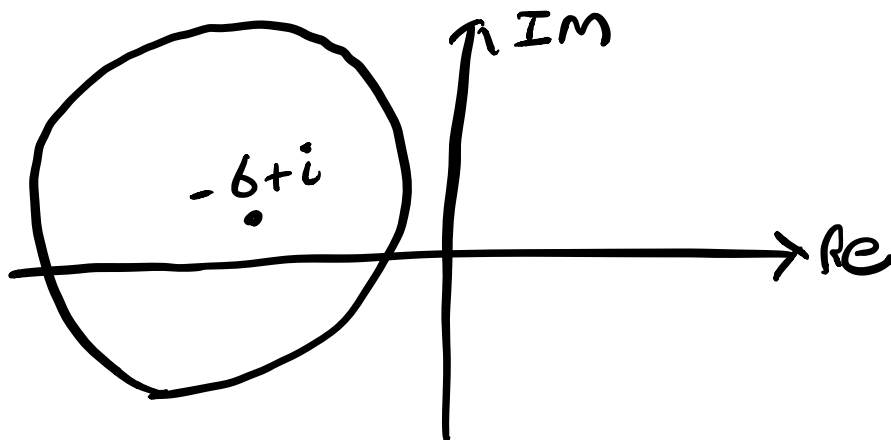
$$b) |z - w| = |w - z|$$

as distance of z from w is same as distance of w from z

$$c) |i - 6 - z| = |z + 6 - i|$$

$$= |z - (-6 + i)| \quad \text{so}$$

$$|z - (-6 + i)| = 5$$



Q9.

$$z = x + yi$$

$$|z - (a + bi)| = |(x - a) + (y - b)i|$$

$$= \sqrt{(x - a)^2 + (y - b)^2}$$

$$\text{so } \sqrt{(x - a)^2 + (y - b)^2} = r$$

$$\text{so } (x - a)^2 + (y - b)^2 = r^2$$

hence why its a circle radius: r
centre: (a, b)

Q10. a)

$$\text{Arg} \left(\frac{z - 3}{z + 2} \right) = \frac{\pi}{4}$$

✓ as given

$$\text{Arg} \left(\frac{z}{w} \right) = \text{Arg } z - \text{Arg } w$$

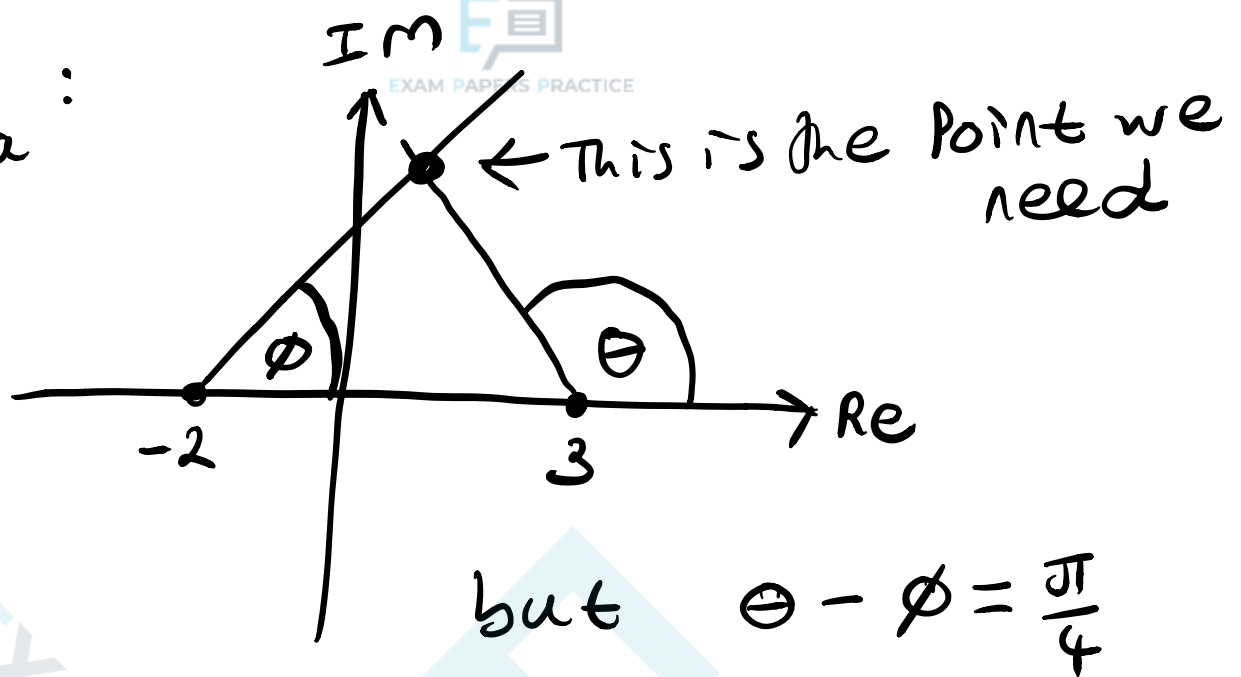
$$\text{so } \text{Arg}(z - 3) - \text{Arg}(z + 2) = \frac{\pi}{4}$$

$$\text{so } \theta - \phi = \frac{\pi}{4}$$

b) so difference between angles

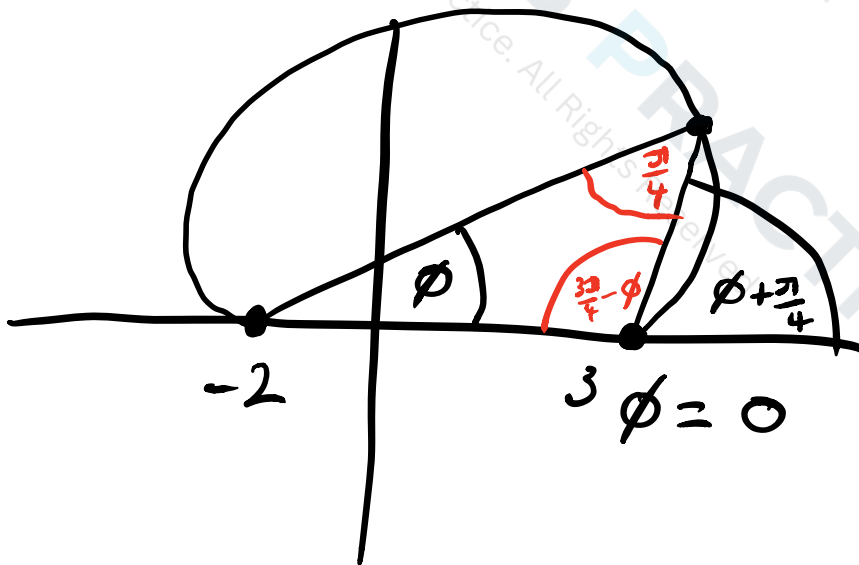
$$\theta, \phi \text{ is } \frac{\pi}{4}$$

rough
idea :



$$\text{or } \Theta = \phi + \frac{\pi}{4}$$

Then for any ϕ we $\Theta = \phi + \frac{\pi}{4}$



So on triangle with point we plot

